TEST CODE **01207020**

FORM TP 2006048

MAY/JUNE 2006

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

BIOLOGY

Paper 02 – General Proficiency

$1\frac{1}{2}$ hours

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. **There are FIVE questions in this booklet. Answer ALL questions.**
2. **You MUST use this answer booklet when responding to the questions. For each question, write your answer in the space provided and return the answer booklet at the end of the examination.**

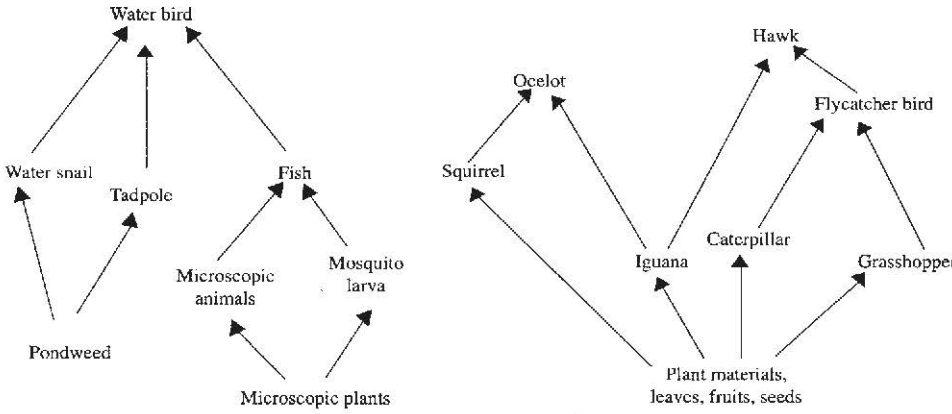
DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

Copyright © 2005 Caribbean Examinations Council®.
All rights reserved.

Answer ALL questions.

Do NOT spend more than 30 minutes on Question 1.

1. Figure 1 below shows an aquatic and a terrestrial food web.



Aquatic food web **Terrestrial food web**

Figure 1: Diagrams of two food webs

(a) Table 1 below contains examples of organisms at different trophic levels.

Complete Table 1 using ONLY the organisms shown in the food webs in Figure 1.

TABLE 1: ORGANISMS AT DIFFERENT TROPHIC LEVELS IN FIGURE 1

Trophic levels	Aquatic food web	Terrestrial food web
	Pond weed	Plant materials, leaves, fruits, seeds
Primary consumer		
	Fish	Flycatcher
Tertiary Consumer		

(6 marks)

(b) Identify TWO similarities between the aquatic and terrestrial food webs.

_____

_____

(2 marks)

GO ON TO THE NEXT PAGE

(c) Describe how data may be collected to construct the **terrestrial** food web in Figure 1. Make a list of the materials and apparatus required and state ONE precaution to be taken in collecting the data.

Materials and apparatus: _____

_____

_____

(2 marks)

Method: _____

_____

_____

_____

(4 marks)

Precaution: _____

_____

(1 mark)

(d) In the space provided below, construct a table to present the data to be collected for constructing the terrestrial food web.

(4 marks)

(e) Suggest TWO ways in which the data to be collected may be summarized or represented, apart from food webs.

_____

_____

(2 marks)

(f) (i) How would the method differ from that described in (c), if you wished to gather data to construct the **aquatic** food web?

_____

_____

_____

_____

(2 marks)

(ii) Explain why the method would be different for collecting data for the aquatic food web.

_____

_____

(2 marks)

(g) Decomposers are important to food webs because they recycle nutrients.

(i) Name TWO types of decomposers that are usually found in food webs.

_____

_____

(2 marks)

(ii) Describe the role decomposers play in the recycling process.

_____

_____

(3 marks)

Total 30 marks

2. Closely examine Figure 2 which represents the major parts of the alimentary canal and associated organs.

(a) (i) Complete the labelling of the boxes in Figure 2. **(3 marks)**

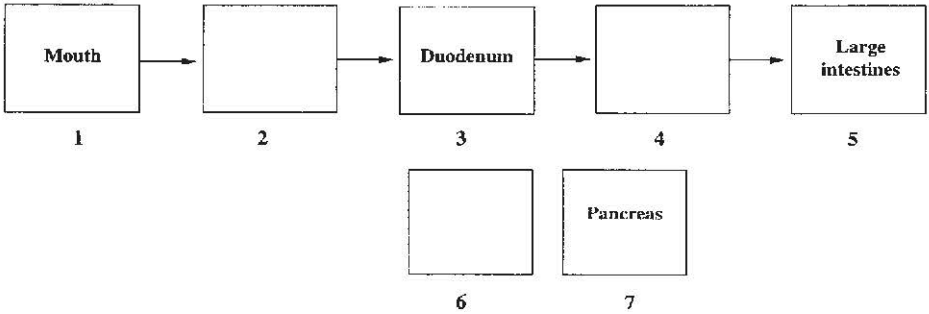


Figure 2. Major parts of the alimentary canal and associated organs

(ii) Place the symbols (P, R, S and T) in the boxes in Figure 2 where the respective activities described below take place:

P: Completion of digestive processes
R: Production of an acidic pH
S: Absorption of large quantities of water
T: Emulsification of fats **(4 marks)**

(iii) Draw ONE arrow on the diagram in Figure 2 to show where digested food is taken. **(1 mark)**

(b) Identify THREE products into which food is converted during the digestive process and give a function of EACH of the products identified.

Digestive product 1: _____

Function: _____

_____

Digestive product 2: _____

Function: _____

_____

Digestive product 3: _____

Function: _____

_____

(6 marks)

Total 14 marks

GO ON TO THE NEXT PAGE

3. Figure 3 is a diagram of a longitudinal section through phloem cells of the vascular bundle of a dicotyledonous plant.

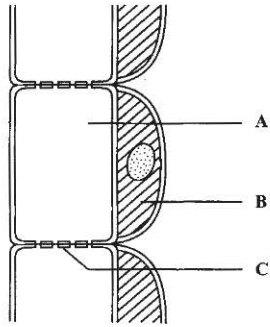


Figure 3. Longitudinal section through phloem cells

(a) (i) Identify the structures labelled A, B and C in Figure 3.

A: _____

B: _____

C: _____

(3 marks)

(ii) State the function of the structure labelled C in Figure 3.

_____

(1 mark)

(iii) State TWO structural differences between xylem and phloem.

_____

_____

_____

(2 marks)

(b) (i) Sometimes roots develop as storage organs. Suggest TWO reasons why a plant may need to store food.

_____

_____

_____

(2 marks)

GO ON TO THE NEXT PAGE

(ii) Figure 4 represents the cross-section of a dicotyledonous root.

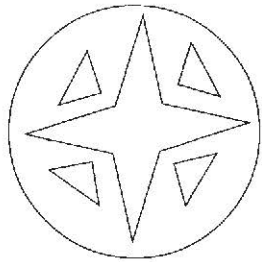


Figure 4. Cross-section of a dicotyledonous root

a) Place an **X** on Figure 4, in the area where food material is likely to be stored. **(1 mark)**

b) Label any THREE structures in Figure 4. **(3 marks)**

(c) Plants manufacture the food they store by the process of photosynthesis.

(i) State TWO environmental factors that are necessary for photosynthesis to occur.

_____

_____

_____

(2 marks)

(ii) How does food manufactured in photosynthesis reach storage areas in the root?

_____

_____

(2 marks)

Total 16 marks

4. Figure 5 shows diagrams of the male and female reproductive organs of humans.

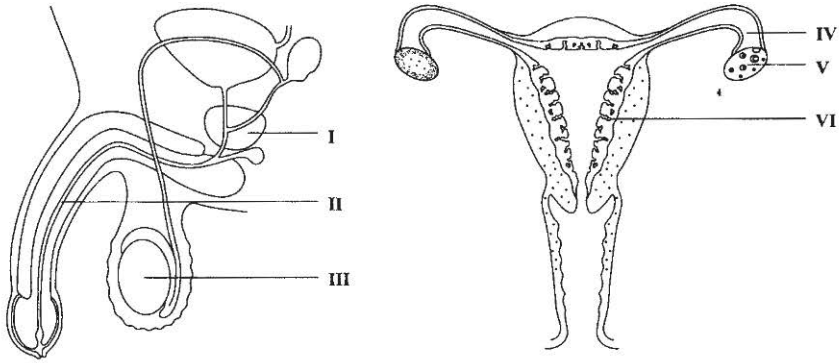


Figure 5. Male and female reproductive organs

(a) (i) Complete Table 2 below by writing the name and function of EACH of the structures labelled I, II, IV and VI.

TABLE 2: NAME AND FUNCTION OF STRUCTURE

	Name of Structure	Function
I		
II		
IV		
VI		

(8 marks)

(ii) Explain why III and V, shown in Figure 5, may be described as having similar functions.

_____

_____

(2 marks)

GO ON TO THE NEXT PAGE

(b) (i) Name the product of fertilization.

_____

(1 mark)

(ii) What becomes of the product of fertilization in

a) humans

_____

_____

b) flowering plants?

_____

_____

(2 marks)

(c) Identify TWO ways in which the male and female reproductive organs illustrated in Figure 5 **function** differently from each other.

_____

_____

_____

(2 marks)

Total 15 marks

5. Figure 6 below represents the offspring produced when two pure-breeding dogs are crossed and their offspring are interbred (offspring from the same litter are bred).

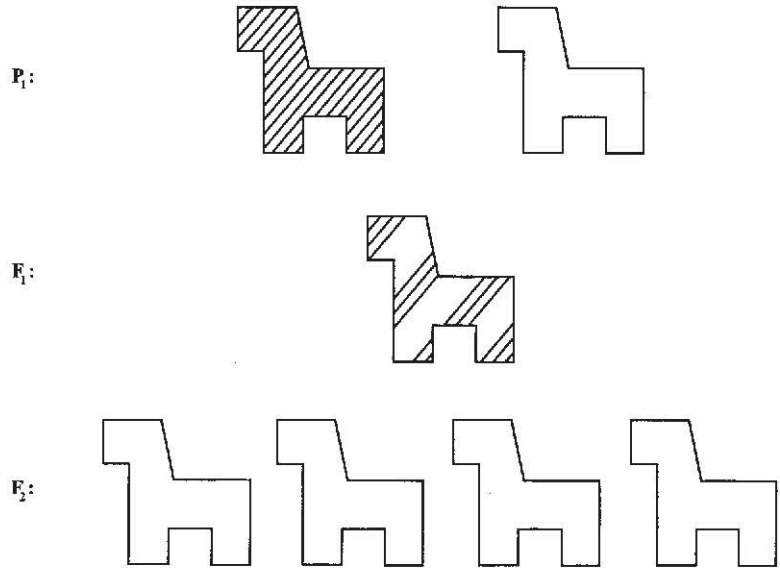


Figure 6. Offspring from two pure-breeding parents

*Note: the diagrams showing coat colour in the F_2 generation are **not** complete.*

(a) (i) Parents (P_1) represent pure-breeding dogs for coat colour. In the space provided below, use appropriate symbols to represent the coat colour for both parents and the F_1 generation.

(3 marks)

(ii) Using appropriate stripes, complete the diagrams of the F_2 generation in Figure 6, to show the coat colours of the offspring when the F_1 generation interbreed.

(4 marks)

GO ON TO THE NEXT PAGE

(b) Coat colour is an example of phenotype.

(i) Explain the meaning of the term 'phenotype'.

(2 marks)

(ii) How is the phenotype of an organism determined?

(2 marks)

(c) (i) Why is genetic variation important?

(2 marks)

(ii) 'Monocropping' – planting of the same type of plants over an extensive area – is sometimes considered risky because of the lack of genetic variation.

Explain why this practice may be considered risky.

(2 marks)

Total 15 marks

END OF TEST